David Weizhong Liu

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Website — LinkedIn — GitHub — Google Scholar

Research Interests

I am currently working on improving long video reasoning in multimodal LLMs through agentic approaches, and I am also very interested in creating both richer and more cost-efficient representations of 3D/4D visual information in LLMs. Broad topics I am interested in include computer vision, agents, robotics, AI for science, and symbolic distillation.

Education

Princeton University, Princeton, NJ — *Aug 2024 – May 2028*
B.S.E. in Computer Science, Minor in Statistics & Machine Learning

Athens Academy, Athens, GA — *Aug 2020 – May 2024*
High School Diploma

Publications & Conferences

1. InFlux++: Real and Synthetic Data for Estimating Dynamic Camera Intrinsics
   Erich Liang, Caleb Kha-Uong, Chinmaya Saran, Sreemanti Dey, **David W Liu**, Junhan Ouyang, Benjamin Zhou, Jia Deng
   European Conference on Computer Vision (ECCV), 2026

2. ViThinker: Active Vision-Language Reasoning via Dynamic Perceptual Querying
   Weihang You, Qingchan Zhu, **David Liu**, Yi Pan, Geng Yuan, Hanqi Jiang
   IEEE International Conference on Multimedia and Expo (ICME), 2026

3. Digital Twin/MARS-CycleGAN: Enhancing Sim-to-Real Crop/Row Detection for MARS Phenotyping Robot Using Synthetic Images
   David Liu*, Zhengkun Li*, Zihao Wu*, Changying Li
   Journal of Field Robotics: Volume 42, Issue 3

4. APP-RUSS: Automated Path Planning for Robotic Ultrasound System
   David Liu, Jerome Charton, Xiang Li, Quanzheng Li
   Presented at IEEE International Symposium on Biomedical Imaging (ISBI), 2024

5. Digital Twin (DT)-CycleGAN: Enabling Zero-Shot Sim-to-Real Transfer of Visual Grasping Models
   David Liu, Yuzhong Chen, Zihao Wu
   IEEE Robotics and Automation Letters: Volume 8, Issue 5
   Presented at IEEE International Conference on Intelligent Robots & Systems (IROS), 2023

6. Hierarchical Optimal Path Planning (HOPP) for Robotic Apple Harvesting
   David W Liu
   International Journal of High School Research: Volume 4, Issue 3

Manuscripts in Preparation/Submission

1. Beyond Visual Inspection: Measurement Interfaces for Video Agents
   David W Liu, Wenhao Chai, Sihan Xu, Enxin Song, Karthik Narasimhan
   Submitting to CVPR 2027

2. Evidence-Derived Answering for Medical Vision-Language Models
   Hanqi Jiang, Junhao Chen, Weihang You, **David W Liu**, Qingchan Zhu, Hellen Luo, Yi Pan, Tiance Wang, Zhengliang Liu, Geng Yuan, Wei Liu, Lei Xing
   Submitting to CVPR 2027

3. Harvest: Conditional Generation of Natural Product Structures
   David Meijer, Allison Judge, **David Liu**, Joseph Clark, Justin J. J. van der Hooft, Marnix H. Medema, Chad Johnston, Michael A. Skinnider
   Manuscript in Preparation

4. Compositional LLM Routing via Subtask Decomposition and Query–Model Embeddings
   David Liu*, Hellen Luo*, Edison Zhu
   Submitting to ACL 2027 Student Research Workshop

*denotes equal contribution

Research Experience

Princeton NLP Group - Undergraduate Researcher *Aug 2026 – Present*

– advised by Wenhao Chai & Prof. Karthik Narasimhan

Ludwig Cancer Research - Undergraduate Research Intern *May 2026 – Present*

– advised by Dr. David Meijer & Prof. Michael Skinnider

Princeton Vision & Learning Lab - Undergraduate Researcher *Oct 2025 – Present*

– advised by Erich Liang & Prof. Jia Deng

United Imaging Intelligence - Research Intern *May 2025 – Aug 2025*

– advised by Dr. Xiao Chen

Harvard University - Research Assistant *Jun 2023 – Oct 2023*

– advised by Dr. Jerome Charton, Prof. Xiang Li, & Prof. Quanzheng Li

University of Georgia - Research Assistant *Jun 2021 – Jul 2024*

– advised by Dr. Zihao Wu & Prof. Changying Li

Industry Experience

Everglades Analytics - Software Engineer *Aug 2026 – Present*

Alexandria AI - Machine Learning Engineer *Aug 2025 – Jan 2026*

Leadership Roles

Princeton COS Council — Academic Chair *Apr 2026 – Apr 2027*

Princeton Club Rugby — Gear Chair *Apr 2025 – Apr 2026*

Athens Academy Robotics Team - Captain *Aug 2022 – May 2024*

Athens Academy Varsity Football - Captain *Aug 2023 - Nov 2023*

Awards

Athens Academy - Valedictorian *Apr 2024*

Georgia High School Association - All-Region Honorable Mention in Football *Dec 2023*

75th Georgia Science and Engineering Fair - Creative Award in Computing *Apr 2023*

75th Georgia Science and Engineering Fair - Innovative Award in Cyber-Physical Systems *Apr 2023*

Georgia Independent Schools Association - 2020 Spelling Bee State Champion *Mar 2020*